

21-4b How Do Wages Affect Labor Supply?

So far, we have used the theory of consumer choice to analyze how a person allocates income between two goods. We can apply the same theory to analyze how a person allocates time. People spend some of their time enjoying leisure and some of it working so they can afford to buy consumption goods. The essence of the time-allocation problem is the trade-off between leisure and consumption.

Consider the decision facing Jasmine, a freelance software designer. Jasmine is awake for 100 hours per week. She spends some of this time enjoying leisure—playing *Fortnite*, watching *The Bachelor*, and reading this textbook. She spends the rest of this time at her computer developing software. For every hour she works developing software, she earns \$50, which she spends on consumption goods—food, clothing, and music downloads. Her hourly wage of \$50 reflects the trade-off Jasmine faces between leisure and consumption. For every hour of leisure she gives up, she works one more hour and gets \$50 of consumption.

Figure 14 shows Jasmine's budget constraint. If she spends all 100 hours enjoying leisure, she has no consumption. If she spends all 100 hours working, she has a weekly consumption of \$5,000 but no time for leisure. If she works a 40-hour week, she enjoys 60 hours of leisure and has weekly consumption of \$2,000.

Figure 14

The Work-Leisure Decision

This figure shows Jasmine's budget constraint for deciding how much to work, her indifference curves for consumption and leisure, and her optimum.

Figure 14 uses indifference curves to represent Jasmine's preferences for consumption and leisure. Here consumption and leisure are the two "goods" between which Jasmine chooses. Because Jasmine always prefers more leisure and more consumption, she prefers points on higher indifference curves to points on lower ones. At a wage of \$50 per hour, Jasmine chooses a combination of consumption and leisure represented by the point labeled "optimum." The optimum is the point on the budget constraint at which Jasmine reaches the highest possible indifference curve, I_2 .

Now consider what happens when Jasmine's wage increases from \$50 to \$60 per hour. Figure 15 illustrates two possible outcomes. In both cases, the budget constraint, shown in the left graphs, shifts outward from BC_1 to BC_2 . In the process, each budget constraint becomes steeper, reflecting the change in relative price: At the higher wage, Jasmine earns more consumption for every hour of leisure that she gives up.

Figure 15

An Increase in the Wage

The two panels of this figure show how a person might respond to an increase in the wage. The graphs on the left show the consumer's initial budget constraint, BC_1 , and new budget constraint, BC_2 , as well as the consumer's optimal choices over consumption and leisure. The graphs on the right show the resulting labor-supply curve. Because hours worked equal total hours available minus hours of leisure, any change in leisure implies an opposite change in the quantity of labor supplied. In panel (a), when the wage rises, consumption rises and leisure falls, resulting in a labor-supply curve that slopes upward. In panel (b), when the wage rises, both consumption and leisure rise, resulting in a labor-supply curve that slopes backward.

Jasmine's preferences, as represented by her indifference curves, determine how her choice regarding consumption and leisure responds to the higher wage. In both panels, consumption rises. Yet the responses of leisure to the wage change are different in the two cases. In panel (a), Jasmine responds to the higher wage by enjoying less leisure. In panel (b), Jasmine responds by enjoying more leisure.

Jasmine's decision between leisure and consumption determines her supply of labor because the more leisure she enjoys, the less time she has left to work. In each panel of [Figure 15](#), the right graph shows the labor-supply curve implied by Jasmine's decision. In panel (a), a higher wage induces Jasmine to enjoy less leisure and work more, so the labor-supply curve slopes upward. In panel (b), a higher wage induces Jasmine to enjoy more leisure and work less, so the labor-supply curve slopes "backward."

At first, the backward-sloping labor-supply curve is puzzling. Why would a person respond to a higher wage by working less? The answer comes from considering the income and substitution effects of a higher wage.

Consider first the substitution effect. When Jasmine's wage rises, leisure becomes more expensive relative to consumption, encouraging Jasmine to substitute away from leisure and toward consumption. In other words, the substitution effect induces Jasmine to work

more in response to higher wages and thus tends to make the labor-supply curve slope upward.

Now consider the income effect. When Jasmine's wage rises, she moves to a higher indifference curve, so she is now better off than before. As long as consumption and leisure are both normal goods, Jasmine will want to use her increased well-being to enjoy both higher consumption and greater leisure. In other words, the income effect induces her to work less and thus tends to make the labor-supply curve slope backward.

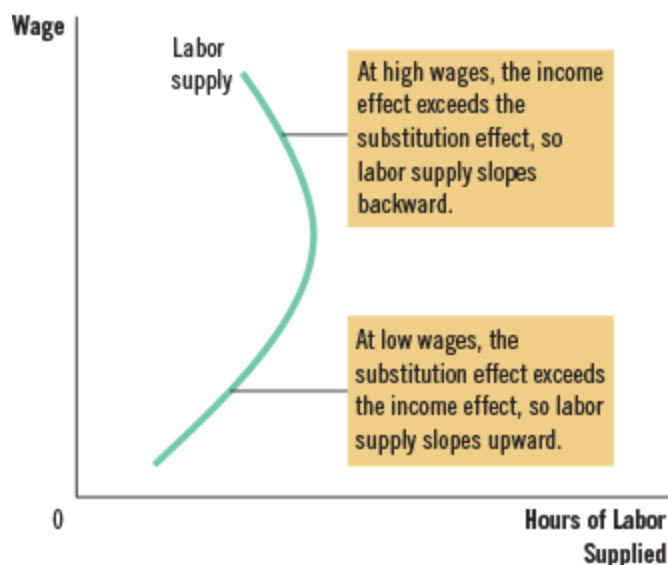
In the end, economic theory does not give a clear prediction about whether an increase in the wage induces Jasmine to work more or less. If the substitution effect is greater than the income effect, she works more. If the income effect is greater than the substitution effect, she works less. The labor-supply curve, therefore, could be either upward- or backward-sloping.

Moreover, the slope of the labor-supply curve need not be the same at all wages. For example, depending on a person's preferences, it is possible for the substitution effect to dominate the income effect at low wages and the income effect to dominate the substitution effect at high wages. In this case, as in [Figure 16](#), the labor-supply curve starts off upward-sloping but then bends backward as the wage increases.

Figure 16

A Backward-Bending Labor-Supply Curve

Here the labor-supply curve slopes upward at low wages because the substitution effect dominates the income effect. But as the wage rises, the income effect starts to dominate the substitution effect, and so the labor-supply curve bends backward.



Case Study: Income Effects on Labor Supply: Historical Trends, Lottery Winners, and the Carnegie Conjecture

The idea of a backward-sloping labor-supply curve might at first seem like a mere theoretical curiosity, but in fact it is not. Evidence indicates that the labor-supply curve, considered over long periods, does indeed slope backward. A hundred years ago, many people worked six days a week. Today, five-day work-weeks are the norm. At the same time that the length of the workweek has been falling, the wage of the typical worker (adjusted for inflation) has been rising.

Here is how economists explain this historical pattern: Over time, advances in technology raise workers' productivity and, thereby, the demand for labor. This increase in labor demand raises equilibrium wages. As wages rise, so does the reward for working. Yet rather than responding to this increased incentive by working more, most workers choose to take advantage of their greater prosperity by enjoying more leisure. In other words, the income effect of higher wages dominates the substitution effect.

Further evidence that the income effect on labor supply is strong comes from a very different kind of data: winners of lotteries. Winners of large prizes in the lottery see large increases in their incomes and, as a result, large outward shifts in their budget constraints. Because the winners' wages have not changed, however, the *slopes* of their budget constraints remain the same. There is, therefore, no substitution effect. By examining the behavior of lottery winners, we can isolate the income effect on labor supply.

The results from studies of lottery winners are striking. Of those winners who win more than \$50,000, almost 25 percent quit working within a year and another 9 percent reduce the number of hours they work. Of those winners who win more than \$1 million, almost 40 percent stop working. The income effect on labor supply of winning such a large prize is substantial.

Similar results were found in a 1993 study, published in the *Quarterly Journal of Economics*, of how receiving a bequest affects a person's labor supply. The study found that a single person who inherits more than \$150,000 is four times as likely to stop working as a single person who inherits less than \$25,000.

This finding would not have surprised the 19th-century industrialist Andrew Carnegie. Carnegie warned that "the parent who leaves his son enormous wealth generally deadens the talents and energies of the son, and tempts him to lead a less useful and less worthy life than he otherwise would." That is, Carnegie viewed the income effect on labor supply to be substantial and, from his paternalistic perspective, regrettable. During his life and at his death, Carnegie gave much of his vast fortune to charity.

"No more 9 to 5 for me."

AP images/John Hagerty

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